

The Dot Product



Computing dot products and angles

- 1 Let $\mathbf{u} = \langle 2, 5 \rangle$ and $\mathbf{v} = \langle -3, 4 \rangle$. Find:
 - a $\mathbf{u} \cdot \mathbf{v}$
 - b $|\mathbf{u}|$ and $|\mathbf{v}|$
 - c the angle between $\mathbf{u}$ and $\mathbf{v}$, to one decimal place
 - 2 Determine whether each pair of vectors is orthogonal:
 - a $\langle 4, -6 \rangle$ and $\langle 3, 2 \rangle$
 - b $\langle 1, 2 \rangle$ and $\langle 2, 1 \rangle$
 - 3 Find the value of k so that $\langle k, 3 \rangle$ is orthogonal to $\langle 2, -4 \rangle$.
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Projections and work

- 4 Find the vector projection of $\mathbf{u} = \langle 6, 2 \rangle$ onto $\mathbf{v} = \langle 3, 0 \rangle$.
 - 5 A force $\mathbf{F} = \langle 8, 5 \rangle$ N moves an object along the displacement $\mathbf{d} = \langle 10, 0 \rangle$ m. Find the work done.
 - 6 A wagon is pulled a distance of 20 m by a rope making an angle of 30° with the ground, with a force of magnitude 50 N. Find the work done, to one decimal place.
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Properties of the dot product

- 7 Let $\mathbf{u} = \langle 3, 1 \rangle$, $\mathbf{v} = \langle -2, 4 \rangle$, $\mathbf{w} = \langle 1, 1 \rangle$. Verify that $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ (the distributive property).
 - 8 Show that $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$ for $\mathbf{u} = \langle 4, -3 \rangle$.
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Angle classification

- 9 Without finding the angle exactly, classify each pair of vectors as forming an acute, right, or obtuse angle, using only the sign of the dot product:
 - a $\langle 3, 5 \rangle$ and $\langle 2, -1 \rangle$
 - b $\langle -1, 4 \rangle$ and $\langle 8, 2 \rangle$
 - c $\langle 2, 2 \rangle$ and $\langle -3, -3 \rangle$

Scalar projection and further applications

- 10 Find the scalar projection of $\mathbf{u} = \langle 5, -2 \rangle$ onto $\mathbf{v} = \langle 3, 4 \rangle$ (the signed length $\text{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|}$), to two decimal places.
- 11 A sled is pulled by a rope at 40° above horizontal with a force of 120 N, moving the sled 15 m horizontally. Find the work done by the horizontal component of the force.

Decomposing a vector into parallel and perpendicular parts

- 12 Decompose $\mathbf{u} = \langle 7, 4 \rangle$ into a component parallel to $\mathbf{v} = \langle 1, 0 \rangle$ and a component perpendicular to $\mathbf{v}$.

Deriving the dot product angle formula

- 13 State the formula relating the dot product to the angle between two vectors, and explain why $\cos \theta = 0$ corresponds to orthogonal vectors.

The Cauchy-Schwarz inequality (preview)

- 14 Explain why $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$ must always hold, using the formula $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos \theta$.

Dot products with variable components

- 15 Find all values of k such that $\langle k, 3 \rangle$ and $\langle k, -3 \rangle$ are orthogonal.
- 16 Find all values of k such that the angle between $\langle 1, k \rangle$ and $\langle 2, 1 \rangle$ is 45° .

A capstone dot product application

- 17 A weather vane experiences wind force $\mathbf{F} = \langle 15, -8 \rangle$ N. The vane can only rotate about a pivot, so only the component of force perpendicular to the vane's current direction $\mathbf{d} = \langle 3, 4 \rangle$ causes it to turn. Find the magnitude of the perpendicular (turning) component of the force, using the fact that the perpendicular component's magnitude equals $\frac{|\mathbf{F} \times \mathbf{d}|}{|\mathbf{d}|}$ in 2D (using the scalar cross product $F_x d_y - F_y d_x$), and separately confirm the parallel component using the dot product.

- 18 Two salespeople's monthly sales across three product categories are given by vectors $\mathbf{s}_1 = \langle 40, 25, 10 \rangle$ and $\mathbf{s}_2 = \langle 15, 30, 45 \rangle$ (units sold). Find $\mathbf{s}_1 \cdot \mathbf{s}_2$ and explain what this dot product does NOT directly tell you about their relative performance (versus what the individual sums would show).